early atmospheres, we can duplicate them in the laboratory, supply some energy, and see which organic molecules are made and in what amounts. Such experiments have over the years proved provocative and promising. But our ignorance of initial conditions limits their relevance.

What we need is a real world whose atmosphere still retains some of those hydrogen-rich gases, a world in other respects something like the Earth, a world in which the organic building blocks of life are being massively generated in our own time, a world we can go to to seek our own beginnings. There is only one such world in the Solar System.* That world is Titan, the big moon of Saturn. It's about 5,150 kilometers (3,200 miles) in diameter, a little less than half the size of the Earth. It takes 16 of our days to complete one orbit of Saturn.

No world is a perfect replica of any other, and in at least one important respect Titan is very different from the primitive Earth: Being so far from the Sun, its surface is extremely cold, far below the freezing point of water, around 180° below zero Celsius. So while the Earth at the time of the origin of life was, as now, mainly ocean-covered, plainly there can be no oceans of liquid water on Titan. (Oceans made of other stuff are a different story, as we shall see.) The low temperatures provide an advantage, though, because once molecules are synthesized on Titan, they tend to stick around: The higher the temperature, the faster molecules fall to pieces. On Titan the molecules that have been raining down like manna from heaven for the last 4 billion years might still be there, largely unaltered, deep-frozen, awaiting the chemists from Earth.

THE INVENTION OF THE TELESCOPE in the seventeenth century led to the discovery of many new worlds. In 1610 Galileo first spied the four large satellites of Jupiter. It looked like a miniature solar system, the little moons racing around Jupiter as the planets were thought by Copernicus to orbit the Sun. It was another blow to the geocentrists. Forty-five years later, the celebrated Dutch physicist Christianus Huygens discovered a moon moving about the planet Saturn and